MMT-Bench

Abstract

MMT-Bench, p.1

> In this study, we present MMT-Bench, a comprehensive benchmark designed to assess LVLMs across massive multimodal tasks requiring expert knowledge and deliberate visual recognition, localization, reasoning, and planning

MMT-Bench, p.1

> MMT-Bench comprises 31, 325 meticulously curated multi-choice visual questions from various multimodal scenarios such as vehicle driving and embodied navigation, covering 32 core meta-tasks and 162 subtasks in multimodal understanding

MCQ-based questions are present, not open-ended.

MMT-Bench, p.1

> Due to its extensive task coverage, MMT-Bench enables the evaluation of LVLMs using a task map, facilitating the discovery of in- and out-of-domain tasks

Extensive range of tasks reveal shortcomings and strengths of proprietary models.

Conclusion

MMT-Bench, p.9

> Our evaluation reveal significant challenges for current LVLMs posed by our MMT-Bench.

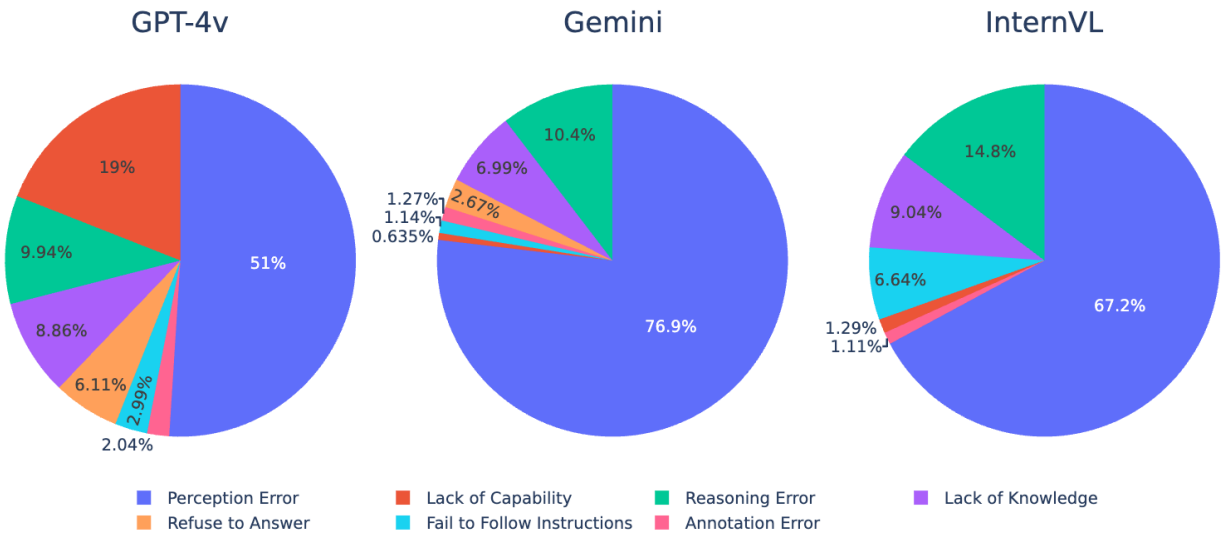
Specific challenge is not mentioned in Conclusion

Tables and figures

Table 1. The comparison between MMT-Bench and existing evaluation benchmarks. MMT-Bench consists of massive samples and multimodal tasks compared with other benchmarks. I, T, V, and P respectively represent image, text, video, and point cloud.

Benchmark	Data Collection					
	# Sample	# Meta-task	# Task	# Modality	Source	Answer Type
SEED-Bench (Li et al., 2023a)	19K	12	12	I + T + V	Annotated	Multi-Choice
MMBench (Liu et al., 2023c)	3K	2	20	I + T	Repurposed	Multi-Choice
MM-VET (Yu et al., 2023)	0.2K	6	N/A	I + T	Repurposed	Multi-Choice
MMMU (Yue et al., 2023b)	11.5K	6	30	I + T	Annotated	Multi-Choice/Open
Tiny LVLM-eHub (Shao et al., 2023)	2.1K	5	42	I + T	Repurposed	Multi-Choice/Open
MMT-Bench	31K	32	162	I + T + V + P	Repurposed	Multi-Choice

Benchmark modalities



Causes of errors in proprietary models - GPT-4v, Gemini, InternVL

Table A7. Abbreviations for tasks requiring multiple images as inputs (part one). Here we also present the designed prompt examples we used for Single-Image LVLMs and Multi-Images LVLMs.

Task Abbreviation	**Task Name**	**Prompt Example for Single Image LVLMs**	**Prompt example for Multiple Image LVLMs**
AQS	action quality assessment	Question: <image><image><image><image> What is the most probable action quality assessment number obtained by the person in the video? Options: A. 35.99 B. 28.0 C. 11.27 D. 44.98	Question: <image><image><image><image> What is the most probable action quality assessment number obtained by the person in the video? Options: A. 35.99 B. 28.0 C. 11.27 D. 44.98

Questions looks very similar to ours

Table A11. Table index of case study figures by meta-task with associated (error) categories for each LVLM.

Case Figure	Meta-task	Subtask	GPT-4V	GeminiProVision	InternVL-Chat
Fig. A5	Visual Recognition	Landmark Recognition	Lack of Knowledge	No Error	No Error
Fig. A6	Object Localization	Camouflaged Object Detection	Lack of Capability	Perception Error	Perception Error
Fig. A7	Pixel-level Recognition	Image Matting	Perception Error	No Error	Perception Error
Fig. A8	OCR	Handwritten Text Recognition	No Error	Perception Error	Perception Error
Fig. A9	Visual Prompt Understanding	Visual Prompt Understanding	No Error	Perception Error, Fail to Follow Instruct	No Error
Fig. A10	Retrieval	Sketch to Image Retrieval	Perception Error	No Error	Perception Error, Reasoning Error
Fig. A11	Counting	Counting by Reasoning	Perception Error	Perception Error	No Error
Fig. A12	Keypoint Detection	Human Keypoint Detection	Refuse to Answer	Perception Error, Fail to Follow Instruct	Fail to Follow Instruct
Fig. A13	Action Recognition	Sign Language Recognition	Lack of Capability	Perception Error	Perception Error
Fig. A14	Visual Hallucination	Exist Hallucination	No Error	Reasoning Error	Perception Error
Fig. A15	Anomaly Detection	Industrial Produce Anomaly Detection	Lack of Knowledge	No Error	Perception Error
Fig. A16	Image-to-Image Translation	Jigsaw Puzzle Solving	No Error	Perception Error	Perception Error
Fig. A17	Visual Summary	Image Captioning Paragraph	Perception Error	No Error	Perception Error
Fig. A18	Intelligence Quotient Test	Ravens Progressive Matrices	No Error	Reasoning Error	Reasoning Error
Fig. A19	Emotional Quotient Test	Scene Emotion Recognition	Perception Error, Reasoning Error	Reasoning Error	No Error
Fig. A20	Visual Grounding	Referring Detection	Perception Error	Perception Error	Fail to Follow Instruct
Fig. A21	Visual Commonsense Reasoning	Whoops	Reasoning Error	Perception Error	Perception Error
Fig. A22	Chart, Doc Understanding	Clock Reading	Perception Error	Perception Error	Perception Error
Fig. A23	Relation Reasoning	Scene Graph Recognition	No Error	Perception Error	No Error
Fig. A24	Meme Understanding	Meme Image Understanding	Perception Error	No Error	No Error
Fig. A25	Multi-Image Analysis	Spot the Diff	No Error	No Error	No Error
Fig. A26	Temporal Understanding	Temporal Ordering	Perception Error	No Error	Perception Error
Fig. A27	Cross-Image Matching	Single Object Tracking	Lack of Capability	Perception Error	Perception Error
Fig. A28	Visual Coding	Equation to Latex	Perception Error	Perception Error	No Error
Fig. A29	Visual Illusion	Color Constancy	Perception Error	No Error	Perception Error
Fig. A30	Image Evaluation Judgement	LVLM Response Judgement	Reasoning Error	No Error	Perception Error
Fig. A31	3D Perception	3D CAD Recognition	Lack of Capability	No Error	No Error
Fig. A32	Emodied Agent	Navigation	Fail to Follow Instruct	Fail to Follow Instruct	Fail to Follow Instruct
Fig. A33	Medical Understanding	Medical Modality Recognition	No Error	No Error	Perception Error
Fig. A34	Autonomous Driving	Traffic Light Understanding	Refuse to Answer	No Error	No Error
Fig. A35	GUI Navigation	Installation	Perception Error	Perception Error	Perception Error
Fig. A36	Discipline Knowledge Reasoning	Art and Design	Lack of Knowledge	Lack of Knowledge	Lack of Knowledge

Error causes:

1. Perception Error
2. Reasoning Error
3. Lack of Knowledge
4. Lack of capability
5. Refuse to answer
6. Fail to follow instructions

Question: How do authors categorize these errors?

Introduction

MMT-Bench, p.1

> However, these benchmarks have limited coverage of multimodal tasks while testing rudimentary capabilities like visual recognition and text-scarce OC

Existing benchmarks have limited coverage of multimodal tasks while testing rudimentary capabilities like visual recognition and text-scarce OCR.

- MMT-Bench is very (I mean very very.....n times) extensive. Large variety in image types including artistic, synthetic, etc.

MMT-Bench, p.2

> The open-source model InternVL-chat has taken a leading position in MMT-Bench, surpassing other closedsource models such as QWen-VL-Plus, GPT-4V, and GeminiProVision.

InternVL-Chat needs to be used in our method

MMT-Bench, p.2

> The comprehensive error analyses conducted on 162 multimodal tasks reveal that top-performing LVLMs such as InternVL-chat, GPT4V, and GeminiProVision are predominantly prone to perception, reasoning, and knowledge errors

MMT-Bench, p.2

> The taxonomy analysis shows that current LVLMs perform well in tasks related to visual recognition and description which are in-domain tasks, yet fall short in tasks related to localization and pixel perception which are out-of-domain tasks

QUESTION: What are out-of-domain and in-domain tasks?

MMT-Bench, p.2

> BLIP2 that does not undergo instruction tuning even outperforms most LVLMs that are tuned by millions of instruction-following data, implying that instructiontuning with data in some tasks even hurts the generalization on other tasks.

MMT-Bench, p.3

> Model performance significantly improves with an increase in size (7B to 13B) for both LLaVA-v1.5 and LLaVA- v1.5-Xtuner. Upgrading LLMs, from InternLM to InternLM2, also enhances the performance of LLaVA

Method

- The data collection pipeline and task setup was done meticuluously. After finalizing 32 meta-tasks, they are further subdivided into several sub-tasks. At the end, after filtering

them with a quality control criteria, 162 sub-tasks remain.
- There are 31, 325 multi-choice questions and 13 input image types.

Experiment

MMT-Bench, p.5

> Evaluation Methods. In MMT-Bench, samples are in a multi-choice format, e.g., 'What is this? Options: (A) Dog (B) Cat'. To extract the choice from LVLMs' responses, we follow OpenCompass' protocol (Contributors, 2023a): 1) Check if the response includes option letters (A/B); 2) Check for option content ('dog'/'cat'); 3) Use ChatGPT for extraction. If these steps fail, we set the model selection as option letter Z to avoid random assignment (Yue et al., 2023a). Accuracy is the primary metric.

The authors use *OpenCompass* protocol. We can also try to use this in our method.

MMT-Bench, p.5

> benchmark poses significant challenges, with even advanced models like InternVL-Chat, GPT-4V and GeminiProVision achieving just 63.4%, 62.0% and 61.6% accuracy, respectively, indicating substantial room for improvement

Authors say ~61% is also bad for proprietary models.

Points used by authors for comparison:

1. Comprehensive challenge of MMT-Bench
   - InternVL-Chat, GPT-4V, GeminiProVision all perform with scores of ~65% indicating room for improvement.
2. Comparison between open-source and close-source LVLMs
   - Prop. models perform better than open-source models
   - But, InternVL-Chat outperforms all other models including closed source.
3. Model scaling:
   - Bigger models perform better

NOTE: The authors combine multiple images into single images for single-image setting. Can we also try this experiment for our setup?

Findings

MMT-Bench, p.7

> Multi-images tasks posed significant challenges to current LVLMs, where the best accuracy achieved by GeminiProVision is only 53.8.

MMT-Bench, p.7

> or Multi-Images LVLMs, providing multiple images as prompts instead of a single image boosted the overall performance on these tasks, demonstrating their capabilities to extract beneficial information from multiple images. For instance, for the task of face retrieval (FR), the performance of GeminiProVision increased from 30.5 to 92.5 when providing multiple images as visual prompts

MMT-Bench, p.7

> To our surprise, BLIP2 achieved significant performance gain when switching to a multi-image prompt setting, especially on tasks like general action recognition (GAR) and video captioning (VC). These results highlight the potential of LVLMs to learn more robust unified representations of multiple modalities

MMT-Bench, p.7

> s shown in Fig 4, perception error stands out as the most common type of error across all models, with GPT- 4V exhibiting a significantly lower perception error rate (51%) compared to GeminiProVision (76.9%) and InternVL (67.2%), indicating its superior performance in perception tasks. Reasoning error emerges as the second most prevalent error type, with InternVL having the highest reasoning error rate (14.8%), followed by GeminiProVision (10.4%) and GPT-4V (9.94%), highlighting the challenges all models face in complex reasoning tasks.

Note: This can serve as further fine-grained error analysis for our work.

Authors also provide lots and lots of qualitative examples